

Key Mathematical Concepts

① Linear Space

- values evenly spaced in linear space.
- Regular Interval
- The difference between consecutive number is constant.
- Example: no of clusters in k-means clustering

$$k = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]$$

use above k value to evaluate clustering performance.



↳ Look for optimal k using metrics (within-cluster sum of squares)

② Logarithmic Space

→ Generates number that are evenly spaced on a logarithmic Space

→ The ratio between consecutive number is constant.

→ Example: Finding optimal learning rate (α)

$$\alpha = [0.00001, 0.0001, 0.001, 0.01, 0.1]$$

↳ Use above α value to evaluate best learning rate.



③ Matrix

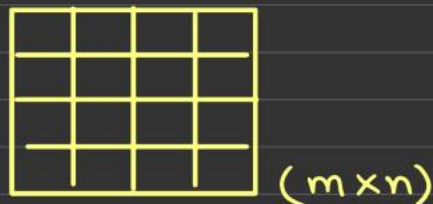
→ In mathematics, a matrix is always 2 dimensional

→ In numpy

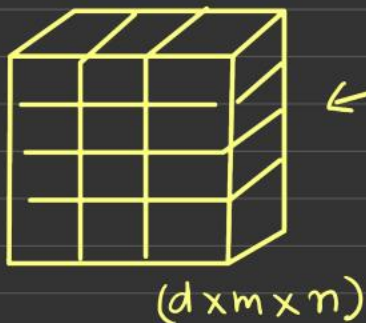
1d array → vector



2d array → Matrix



3d array → Tensor
or higher



→ But today, our focus is on Matrix (2d Array)

2d Array
[Matrix]



$\begin{bmatrix} 5, 10, 15, \\ 2, 4, 6, \\ 5, 7, 2 \end{bmatrix}$

numpy
Style

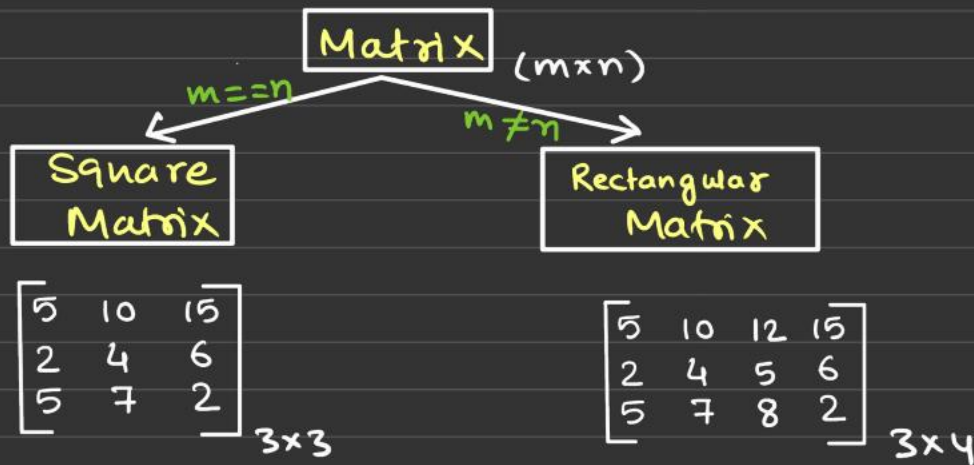


$\begin{bmatrix} 5 & 10 & 15 \\ 2 & 4 & 6 \\ 5 & 7 & 2 \end{bmatrix}$

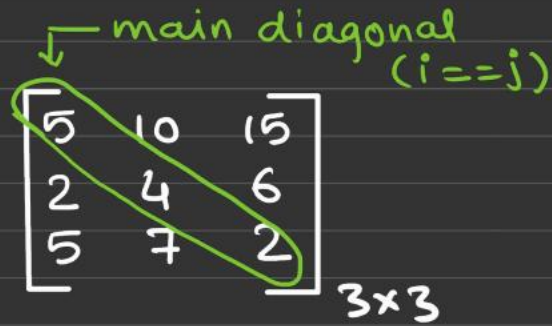
Mathematical

→ 2 dimension
→ Shape → (3 x 3)
→ m = n, so Square
Matrix

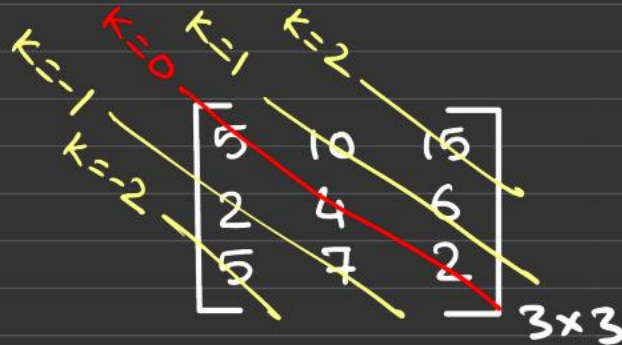
→ Generally matrix can be a Square or Rectangular based on Shape.



* Diagonal of a Matrix



→ diagonal parallel to main diagonal is identified using offset k



→ we will look into some special types of matrix which are only defined for square matrices.

① Diagonal Matrix

$$\begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix}_{3 \times 3}$$

② Identity Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

③ Upper Triangular matrix ↳ Zero below main diagonal

$$\begin{bmatrix} 5 & 10 & 15 \\ 0 & 4 & 6 \\ 0 & 0 & 2 \end{bmatrix}_{3 \times 3}$$

④ Lower Triangular Matrix ↳ Zero above main diagonal

$$\begin{bmatrix} 5 & 0 & 0 \\ 2 & 4 & 0 \\ 5 & 7 & 2 \end{bmatrix}_{3 \times 3}$$

- In Pure Mathematics, both diagonal and Identity matrix are defined for square matrices.
- Even a triangular matrix must be a square matrix.
- But numpy allows above matrices for rectangular matrix as well for programmer's convenience.

Random Number Generation

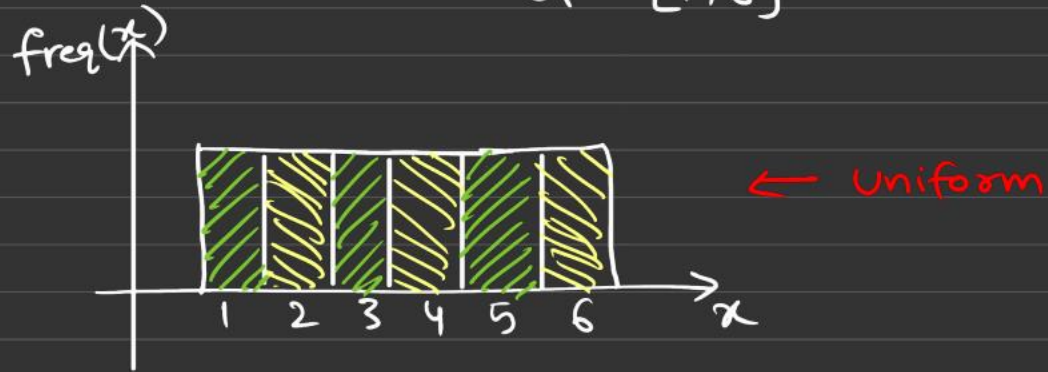
- `np.random.rand()` → Uniform distribution
- `np.random.randn()` → Normal distribution

Ex1: Select a random integer in your mind from $[1, 10]$

↳ Every no. has equal probability of selection = $1/10$

Ex2: Roll a dice 1 times → Random value $[1, 6]$

Roll one dice 10,000 times $\rightarrow$ and plot occurrence of $[1, 6]$



Example 3: Select a Sample of 10,000 adults from India and plot the frequency v/s height graph.

